William hyatt 7th edition drill problems

Electromagnetic Theory (University of Engineering and Technology Lahore)

DRILL PROBLEMS : CHAPTER 2

D2.1

(a) $\mathbf{R_{AB}} = (5+6)\,\mathbf{a_x} + (8-4)\,\mathbf{a_y} + (-2-7)\,\mathbf{a_z} = 11\mathbf{a_x} + 4\mathbf{a_y} - 9\mathbf{a_z}$

(b) $R_{AB} = \sqrt{11^2 + 4^2 + 9^2} = 14.76$ m

(c) $F_{BA} = \frac{(-20\times10^{-6})(50\times10^{-6})}{4\pi\frac{(10^{-9})}{36\pi}(14.76^2)}\,a_{ba} = -0.0413\,\frac{(-11a_x - 4a_y + 9a_z)}{14.76} = 30.78a_x + 11.195a_y - 25.18a_z$ mN

(d) $F_{BA} = \frac{(-20\times10^{-6})(50\times10^{-6})}{4\pi\times8.854\times10^{-12}(14.76^2)}\,a_{ba} = -0.04125\,\frac{(-11a_x - 4a_y + 9a_z)}{14.76} = 30.74a_x + 11.18a_y - 25.15a_z$ mN

D2.2

(a)

$$\boldsymbol{r} - \boldsymbol{r_A} = -25a_x + 30a_y - 15a_z,\ |\boldsymbol{r} - \boldsymbol{r_A}| = 41.43$$
$$\boldsymbol{r} - \boldsymbol{r_B} = 10a_x - 8a_y - 12a_z,\ |\boldsymbol{r} - \boldsymbol{r_B}| = 17.54$$
$$\boldsymbol{E_A} = -1.57\ \frac{(-25a_x + 30a_y - 15a_z)}{41.43} = 9480a_x - 11300a_y + 5600a_z$$
$$\boldsymbol{E_B} = 14.61\ \frac{(10a_x - 8a_y - 12a_z)}{17.54} = 83300a_x - 66600a_y - 99900a_z$$
$$\boldsymbol{E_T} = \boldsymbol{E_A} + \boldsymbol{E_B} = 92.48a_x - 77.9a_y - 94.3a_z\ \frac{kV}{m}$$

(b)

$$\boldsymbol{r} - \boldsymbol{r_A} = -10a_x + 50a_y + 35a_z,\ |\boldsymbol{r} - \boldsymbol{r_A}| = 61.84$$
$$\boldsymbol{r} - \boldsymbol{r_B} = 25a_x + 12a_y + 38a_z,\ |\boldsymbol{r} - \boldsymbol{r_B}| = 47.04$$
$$\boldsymbol{E_A} = -7050\ \frac{(-10a_x + 50a_y + 35a_z)}{61.84} = 1140a_x - 5700a_y - 3990a_z$$
$$\boldsymbol{E_B} = 20300\ \frac{(25a_x + 12a_y + 38a_z)}{47.04} = 10700a_x + 5180a_y + 16400a_z$$
$$\boldsymbol{E_T} = \boldsymbol{E_A} + \boldsymbol{E_B} = 11.84a_x - 0.52a_y + 12.41a_z\ \frac{kV}{m}$$

D2.3

(a) **Sum** $= 2 + 0 + \frac{2}{5} + 0 + \frac{2}{17} + 0 = 2.517$

(b) **Sum** $= \frac{1.1}{11.18} + \frac{1.01}{22.62} + \frac{1.001}{46.87} + \frac{1.0001}{89.44} = 0.1755$

EE08.SOLUTIONS

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D2.4

(a) $Q = \int_{vol} \rho_v \, dv = \int_{-0.2}^{-0.1} \int_{-0.2}^{-0.1} \int_{-0.2}^{-0.1} \frac{1}{x^3y^3z^3} dxdydz + \int_{0.1}^{0.2} \int_{0.1}^{0.2} \int_{0.1}^{0.2} \frac{1}{x^3y^3z^3} dxdydz$

$= -\frac{1}{8\, x^2|_{-02}^{-0.1} y^2|_{-02}^{-0.1} z^2|_{-02}^{-0.1}} - -\frac{1}{8\, x^2|_{0.1}^{0.2} y^2|_{0.1}^{0.2} z^2|_{0.1}^{0.2}} = \frac{1}{8\times(0.03)} - \frac{1}{8\times(0.03)} = 0$

(b) $\int_0^{\pi} \int_0^{0.1} \int_2^{4} \rho^3 z^2 \sin 0.6\varphi \, dz d\rho d\varphi = (-\frac{\cos 0.6\varphi}{0.6})\Big|_0^{\pi} (\frac{\rho^4}{4})\Big|_0^{0.1} (\frac{z^3}{3})\Big|_2^{4} = 1.018\, mC$

(c) $\int_0^{\infty} \int_0^{2\pi} \int_0^{2\pi} e^{-2r} \sin\theta \, d\varphi \, d\theta dr = (-\frac{e^{-2r}}{2})\Big|_0^{\infty} (\cos\theta)|_0^{2\pi} (\varphi)|_0^{2\pi} = -6.28\, C$

D2.5

(a) **E** = $2 \times \frac{5\times10^{-9}}{2\pi\varepsilon_o(4)} \boldsymbol{a_z} = 44.95 \, \frac{V}{m}$

(b) **E**$_x$ = $\frac{5\times10^{-9}}{2\pi\varepsilon_o(5)} \frac{(3\boldsymbol{a_y}+4\boldsymbol{a_z})}{5} = 10.788\boldsymbol{a_y} + 14.384\boldsymbol{a_z} \, \frac{V}{m}$, **E**$_y$ = $\frac{5\times10^{-9}}{2\pi\varepsilon_o(4)} \boldsymbol{a_z} = 22.4775\, \boldsymbol{a_z} \, \frac{V}{m}$

E = **E**$_x$ + **E**$_y$ = $10.788\boldsymbol{a_y} + 36.86\boldsymbol{a_z} \, \frac{V}{m}$

D2.6

i) Electric field due to 3 nC/m^2: **E**$_1$ = $\frac{3\times10^{-9}}{2\varepsilon_o} \boldsymbol{a_N} = 169.5\, \boldsymbol{a_z}$

ii) Electric field due to 6 nC/m^2: **E**$_2$ = $\frac{6\times10^{-9}}{2\varepsilon_o} \boldsymbol{a_N} = 338.8\, \boldsymbol{a_z}$

iii) Electric field due to -8 nC/m^2: **E**$_3$ = $\frac{-8\times10^{-9}}{2\varepsilon_o} \boldsymbol{a_N} = -451.76\, \boldsymbol{a_z}$

According to the direction of point relative to normal:

(a) **E = - E$_1$ - E$_2$ - E$_3$ =** $-56.6\, \boldsymbol{a_z} \, \frac{V}{m}$

(a) **E = E$_1$ - E$_2$ - E$_3$ =** $283\, \boldsymbol{a_z} \, \frac{V}{m}$

(a) **E = E$_1$ + E$_2$ - E$_3$ =** $961\, \boldsymbol{a_z} \, \frac{V}{m}$

(a) **E = E$_1$ + E$_2$ + E$_3$ =** $56.6\, \boldsymbol{a_z} \, \frac{V}{m}$

EE08.SOLUTIONS

D2.7

(a) $\frac{E_y}{E_x} = \frac{dy}{dx} \quad \rightarrow \quad \frac{4x^2}{y^2} \times \frac{y}{8x} = \frac{dy}{dx} \quad \rightarrow \quad \int x dx = -\int 2y dy \quad \rightarrow \quad x^2 + 2y^2 = c$

Put $\mathrm{x} = 1$ and $\mathrm{y} = 2$ to get $c = 33$ giving: $\boldsymbol{x^2 + 2y^2 = 33}$

(b) $\frac{E_y}{E_x} = \frac{dy}{dx} \quad \rightarrow \quad \frac{y(5x+1)}{x} = \frac{dy}{dx} \quad \rightarrow \quad \frac{1}{5} \times \frac{5x+1-1}{5x+1} dx = y dy \quad \rightarrow \quad 0.4x - 0.08 \ln(5x + 1) + c = y^2$

Put $\mathrm{x} = 1$ and $\mathrm{y} = 2$ to get $c = 15.74$ giving: $\boldsymbol{0.4x - 0.08 \ln(5x + 1) + 15.74 = y^2}$

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DRILL PROBLEMS 3

D3.1

(a) Evaluate the triple volume integral to find the total volume enclosed by the portion of sphere / surface and then just multiply it with the given charge to find the total change within it:

$$\int_0^{0.26}\int_0^{\frac{\pi}{2}}\int_0^{\frac{\pi}{2}} r^2 sin\theta \; d\theta d\phi dr \;\times\; q = \frac{1}{8} \times q = 7.5\mu C$$

(b) This surface encloses the whole charge q, so answer is 60 μC

(c) Only the upper half of the flux lines pass through the plane at z = 26 cm, so D = 0.5 x 60 = 30 μC

D3.2

(a) $E = \frac{kQ}{r^2}\frac{4a_x - 6a_y + 12a_z}{\sqrt{16+36+144}} = 0.72a_x - 1.08a_y + 2.16a_z \; \frac{MV}{m}$

$so, D = \varepsilon_o E = 6.38a_x - 9.56a_y + 19.125a_z \; \frac{\mu C}{m^2}$

(b) $E = \frac{20}{2\pi\varepsilon_o.45}\frac{-3a_y + 6a_z}{\sqrt{45}} = -23.97a_y + 47.94a_z \; \frac{MV}{m}$

$so, D = \varepsilon_o E = -212a_y + 424a_z \; \frac{\mu C}{m^2}$

(c) $E = \frac{120}{2\varepsilon_o}a_z = \frac{60}{\varepsilon_o}a_z \; \frac{\mu V}{m}, so\ D = \varepsilon_o E = 60a_z \; \frac{\mu C}{m^2}$

D3.3

(a) $E = \frac{D}{\varepsilon_o} = 33.88r^2 a_r$, so at P: $E = 33.88(2)^2 a_r = 135.5a_r$

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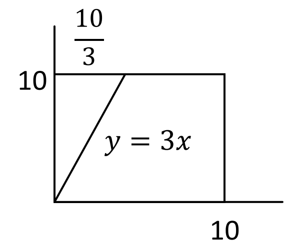
(b) $Q =$
$\oint D.ds = \int_0^{2\pi}\int_0^{\pi} a^2 sin\theta\ d\theta d\phi a_r\ \times\ 0.3r^2 a_r\ =\ 24.3\int_0^{2\pi} -cos\theta|_0^{\pi}\ d\phi =$
$48.6\int_0^{2\pi} d\phi = 305.208\ nC$

(c) On same steps: $Q\ = 76.8\int_0^{2\pi} -cos\theta|_0^{\pi} d\phi = 964.608$ μC

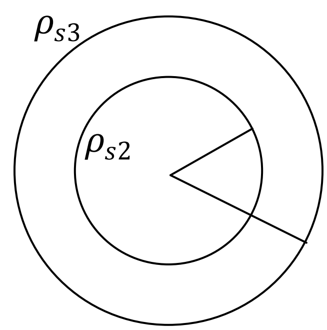
D3.4



(a) $Q\ = Q_1 + Q_2 = 0.243\ \mu C$
(b) $Q\ = length \times \rho_L\ = 31.4\ \mu C$

(c) Area $= length\ of\ line\ (hypotenuse) \times width\ across\ z = 10.53 \times 10$
So, $Q\ = area \times \rho_A\ = 10.53\ \mu C$

D3.5



(a) $D = \frac{0.25}{4\pi(0.005)^2} = 795.77\ \mu C$
(b) $Q_2\ = 4\pi(0.01)^2 \times \rho_{s2}\ = 2.51\ \mu C,$
$so, D = \frac{0.25+2.51}{4\pi(0.015)^2} = 977\ \mu C$
(c) $Q_3\ = 4\pi(0.018)^2 \times \rho_{s3}\ = -2.44\ \mu C,$
$so, D = \frac{0.25+2.51-2.44}{4\pi(0.025)^2} = 40.74\ \mu C$
(d) $D = \frac{0.25+2.51-2.44+4\pi(0.03)^2 \times \rho_s}{4\pi(0.035)^2} = 0$, so $\rho_{s4}\ =$ -28.29 μC

D3.6

(a) $\psi = \oint D.ds = \int_1^3\int_0^2 16x^2yz^3dxdy =\ \int_1^3\int_0^2 16x^2y.8dxdy = 1365\ pC$

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(b) $E = \frac{D}{\varepsilon_o} and\ evaluate\ it\ at\ P$

(c) $Evaluate\ the\ formula\ (8)\ at\ P\ and\ \Delta V = 10^{-12} m^3$

D3.7

$$Evaluate\ the\ corresponding\ formulae\ for\ \text{div}\,\mathbf{D}\ (i.e. 15, 16\ \&\ 17)\ at\ the\ given\ points\ (P)$$

D3.7

$$Take\ \text{div}\,\mathbf{D}\ (as\ stated\ by\ Maxwell's 1st\ equation)\ to\ get\ the\ expressions\ for\ \rho_v, with\ proper\ trignometric\ manipulation$$

D3.8

R.H.S:

$$\nabla.\text{D} = 12sin\frac{\varphi}{2} - 0.75sin\frac{\varphi}{2} = 11.25\ sin\frac{\varphi}{2}$$

$$11.25\int_0^2\int_0^\pi\int_0^5 sin\frac{\varphi}{2}\ \rho dz d\phi d\rho = 11.25 \times 20 = 225$$

L.H.S:

$$\oint D.ds = \int_0^\pi\int_0^5 (\boldsymbol{D})_{\rho=2}\rho dz d\varphi a_\rho + \int_0^2\int_0^5 (\boldsymbol{D})_{\varphi=0} dz d\rho a_\varphi + \int_0^2\int_0^5 (\boldsymbol{D})_{\varphi=\pi} dz d\rho(-a_\varphi)$$

Now,

$$(\boldsymbol{D})_{\rho=2} = 12sin\frac{\varphi}{2}a_\rho$$

$$\int_0^2\int_0^5 (\boldsymbol{D})_{\varphi=\pi} dz d\rho a_\varphi \ = 0$$

$$so, \oint D.ds = 24\int_0^\pi\int_0^5 \ sin\frac{\varphi}{2} dz d\varphi \ + \int_0^2\int_0^5 \ (\boldsymbol{D})_{\varphi=0} dz d\rho a_\varphi$$

$$= 24\left(-2cos\frac{\varphi}{2}\right)|_0^\pi \times (z)|_0^5 - 1.5\left(\frac{\rho^2}{2}\right)|_0^2 \times (z)|_0^5$$

$$= -48\ (0 - 1)(5) - 15$$

$$= 225$$

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EMT Drill Problems
Chapter 4.

D-4.1

$$E = \frac{1}{z^2}\left(8xyz\vec{a}_x + 4x^2z\vec{a}_y - 4x^2y\vec{a}_z\right) \text{ V/m}$$

$|\vec{dL}| = 2\mu m$, $Q = 6\text{-nC}$

$P(2,-2,3)$

a) $a_L = -\frac{6}{7}a_x + \frac{3}{7}a_y + \frac{2}{7}a_z$

$$\vec{dL} = \vec{a}_L \cdot |\vec{dL}|$$
$$= \frac{-12}{7}a_x + \frac{6}{7}a_y + \frac{4}{7}a_z \quad \mu m$$

$$dW = -Q\,E\cdot dL$$
$$= -60\times10^{-9} \times \frac{1}{z^2}\left[\frac{-12}{7}\times 8xyz + \frac{6}{7}\times 4x^2z - \frac{4}{7}\times 4x^2y\right]$$
$$= -60\times10^{-9}\times10^{-6}\left[\frac{-96}{7}xyz + \frac{24}{7}x^2z - \frac{16}{7}x^2y\right]\times\frac{1}{9} \text{ Joules.}$$

but $P(2,-2,3) \Rightarrow x=2,\ y=-2,\ z=3$

$$= \frac{-60\times10^{-15}}{9}\left[164.57 + 41.14 + 18.2857\right]$$
$$= -149.1 \text{ fJ}$$

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b).

$$\vec{dL} = \vec{a}_L \cdot |\vec{dL}|$$
$$= \frac{12}{7}a_x - \frac{6}{7}a_y - \frac{4}{7}a_z$$

$$dW = -Q\,E\cdot dL$$
$$= -60\times10^{-9}\times\frac{1}{z^2}\left[\frac{12}{7}\times 8xyz - \frac{6}{7}\times 4x^2z + \frac{4}{7}\times 4x^2y\right]\mu$$
$$= \frac{-60\times10^{-9}}{9}\times\left[-164.57 - 41.14 - 18.2857\right]$$
$$= 149.1 \text{ fJ}$$

c) $\vec{dL} = \frac{6}{7} a_x + \frac{12}{7} a_y$

$$E \cdot dL = \frac{1}{z^2}\left[\frac{48}{7} xyz + \frac{48}{7} x^2 z\right]$$

$$= \frac{1}{9}\left[\frac{-576}{7} + \frac{576}{7}\right] = 0$$

$\Rightarrow \quad dW = 0$

D: 4-2

$Q = 4C \quad ; \quad B(1,0,0) \; , \; A(0,2,0)$

path: $y = 2 - 2x$
$z = 0$

a) $\vec{E} = 5a_x \; V/m$

$$W = -Q \int_B^A E \cdot dL$$

$$= -4 \int_B^A 5\, dx$$

where $dL = dx\vec{a}_x + dy\vec{a}_y + dz\vec{a}_z$

$$= \cancel{-20 \frac{x^2}{2}\Big|_B^A} \Rightarrow -20x \Big|_B^A$$

$$= \cancel{-10(0-1)} \Rightarrow -20(0-1)$$

$$= 20J$$

b) $\vec{E} = 5x a_x \; V/m$

$$W = -Q \int_B^A E \cdot dL$$

$$= -4 \times \int_B^A 5x\, dx$$

$$= -20 \times \frac{x^2}{2}\Big|_B^A = -10(0-1)$$

$= 10$ Joules

c) $5x a_x + 5y a_y = \vec{E}$

$$W = -Q\int_B^A E\cdot dL = -4\int_B^A (5x\,dx + 5y\,dy)$$

$$= -20\left[\frac{x^2}{2} + \frac{y^2}{2}\right]_B^A$$

$$= -10\left((0-1) + (4-0)\right)$$

$$= -10(-1+4) = -30 \text{ Joules}$$

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D:4-3

$E = y\vec{a}_x$ V/m www.ecuet.net.tc

$Q = 3\text{-}C$

a) $$W = -Q\int_B^A E\cdot dL$$

$$= -3\left(\int_{(1,3,5)}^{(2,3,5)} E\cdot dL + \int_{(2,3,5)}^{2,0,5} E\cdot dL + \int_{(2,0,5)}^{(2,0,3)} E\cdot dL\right)$$

where $E\cdot dL = (y a_x)\cdot(dx a_x + dy a_y + dz a_z)$
$= y\,dx$

$$W = -3\left[\int_{A_1} y\,dx + \int_{A_2} y\,dx + \int_{A_3} y\,dx\right]$$

As x is only varying in the case of A_1 so $dx = 0$ for A_2, A_3.

$$W = -3\int_{A_1} y\,dx$$ as y is constant for $A_1 \Rightarrow y = 3$.

$$= -3\int_{A_1} 3\,dx = -9\,x\Big|_{A_1} =$$

$$= -9(2-1) = -9J$$

b):

$$W = -Q \times \left(\int_{(1,3,5)}^{(1,3,3)} E \cdot dL + \int_{(1,3,3)}^{(1,0,3)} E \cdot dL + \int_{(1,0,3)}^{(2,0,3)} E \cdot dL \right).$$

$E \cdot dL = y\,dx$

as $dx \neq$ for the third integral only ; so the first two integrals are zero.

so

$$W = -3 \int_{(1,0,3)}^{(2,0,3)} y\,dx \qquad y = \text{constant} = \cancel{3}\ 0$$

$$= -9 \times 0 \int dx$$

$$= 0 \text{ Joules}$$

D 4.4):

$$E = 6x^2 a_x + 6y a_y + 4 a_z \quad V/m.$$

a)

$$M(2,6,-1) \quad ; \quad N(-3,-3,2).$$

$$V_{AB} = -\int_B^A E \cdot dL$$

$$= -\int_{(-3,-3,2)}^{(2,6,-1)} \cancel{6x^2 + 6y^2 + 4}\ 6x^2 dx + 6y\,dy + 4dz$$

$$= -\left(2x^3 + 3y^2 + 4z\right)\Big|_{(-3,-3,2)}^{(2,6,-1)}$$

$$= -\left(2\left(2^3 - (-3)^3\right) + 3\left(6^2 - (-3)^2\right) + 4(-1-2)\right)$$

$$= -\left(70 + 81 + \cancel{4} - 12\right)$$

$$= -139 V$$

b) $V_{MQ} = V_M - V_Q$

$\Rightarrow V_M = V_{MQ} + V_Q$

so finding V_{MQ}

$$V_{MQ} = -\int_Q^M E \cdot dL = -\int_Q^M 6x^2dx + 6y\,dy + 4dz$$

$$= -(2x^3 + 3y^2 + 4z)\Big|_{(4,-2,-35)}^{(2,6,-1)}$$

$$= -\left(2(2^3 - 4^3) + 3(6^2 - (-2)^2) + 4(-1+35)\right)$$

$$= -\left[-112 + 96 + 136\right]$$

$$= -120V$$

but given that $V_Q = 0$

$\Rightarrow V_M = V_{MQ} = -120$ volts.

c) $V_{NP} = V_N - V_P$

$V_N = V_{NP} + V_P$

$= V_{NP} + 2 \quad \because V_P = 2$

$$V_{NP} = -\int_P^N E \cdot dL$$

$$= -(2x^3 + 3y^2 + 4z)\Big|_{(1,2,-4)}^{(-3,-3,2)}$$

$$= -\left(2((-3)^3 - 1) + 3(3^2 - 2^2) + 4(2+4)\right)$$

$$V_{NP} = -(-56 + 15 + 24) = 17$$

$\Rightarrow V_N = 17 + 2 = 19$ volts.

D:4.5 :

$Q = 15nC$ at origin

$P_1 \; (-2, 3, -1)$

a) $V = 0$ at $Q(6,5,4)$

As we know

$$V_{AB} = \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{r_A} - \frac{1}{r_b} \right]$$

is the voltage of point A with refrence to point B.

$$r_A = \sqrt{2^2 + 3^2 + 1} = \sqrt{14}$$

$$r_B = \sqrt{6^2 + 5^2 + 4^2} = \sqrt{77}$$

$$V_{AB} = \frac{15 \times 10^{-9}}{4 \times \pi \times \epsilon_0} \left[\frac{1}{\sqrt{14}} - \frac{1}{\sqrt{77}} \right]$$

$$V_{PQ} = 20.67V$$

as

$$V_{AB} = V_A - V_B$$

or $V_{PQ} = V_P - V_Q$ $\quad \because V_B = 0$ given

or $\because V_Q = 0$ "

$$\Rightarrow V_P = 20.67V$$

b) $V = 0$ infinity

$$r_A = \sqrt{14}$$

$$r_b = \infty$$

$$\Rightarrow V_{PQ} = \frac{Q}{4\pi\epsilon_0} \times \left[\frac{1}{\sqrt{14}} - \frac{1}{\infty} \right]$$

$$= 36 \text{ volts}$$

c) $V_Q = 5V$ at $Q(2,0,4)$

$$r_a = \sqrt{14}$$

$$r_b = \sqrt{2^2 + 4^2} = \sqrt{20}$$

$$V_{PQ} = \frac{Q}{4\pi\epsilon_0}\left[\frac{1}{\sqrt{14}} - \frac{1}{\sqrt{20}}\right].$$

$$= 5.885 .$$

but $\quad V_{PQ} = V_P - V_Q .$

$\Rightarrow \quad V_P = V_{PQ} + V_Q$

$$= 5.885 + 5 = 10.885 \text{ volts}$$

D: 4.6

reference $\longrightarrow$ infinity

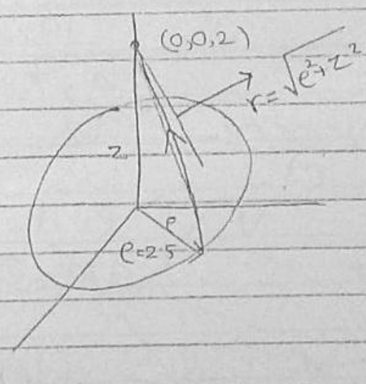
$P(0,0,2)$

a) $\quad \rho_L = 12\,nC/m$

$\rho = 2.5m \quad , z = 0 .$

~~As we know~~ ~~(Eqn 19)~~

$\cancel{V = \int \frac{\rho_L \, dL}{4\pi\epsilon_0 \rho}}$



$$V = \int_0^{2\pi} \frac{\rho_L}{4\pi\epsilon_0 r} dL$$

$$= \int_0^{2\pi} \frac{\rho_L}{4\pi\epsilon_0 r}\left(d\rho\, a_\rho + \rho\, d\phi\, a_\phi + dz\, a_z\right)$$

as $\rho = 2.5$ = constant $\Rightarrow d\rho = 0 .$

as $z = 0$ = " $\Rightarrow dz = 0 .$

so we have

$$= \int_0^{2\pi} \frac{\rho_L}{4\pi\epsilon_0 r} \rho\, d\phi$$

where

$r = \sqrt{\rho^2 + z^2}$

$= \sqrt{2.5^2 + 2^2} = 3.2$

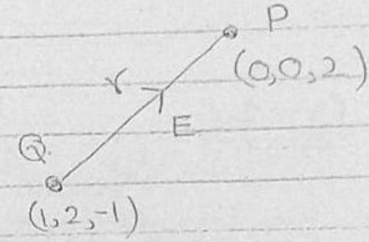
$$= \int_0^{2\pi} \frac{12 \times 10^{-9}}{4 \times \pi \times \epsilon_0 \times 3.2} \times 2.5\, d\phi$$

$$= 84.25 \int_0^{2\pi} d\phi$$

$$= 84.25 \left[\phi \right]_0^{2\pi}$$

$$= 84.25 \times 2\pi = 529.41 \text{ volts}$$

b)- point charge $Q = 18nC$
at $Q(1, 2, -1)$



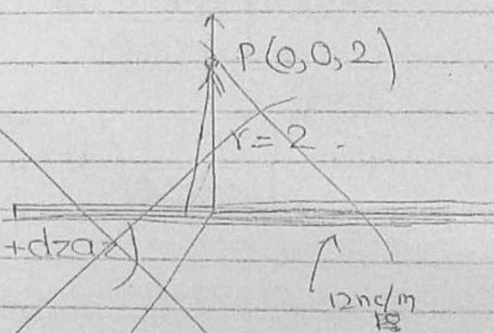
$$V = \frac{Q}{4\pi\epsilon_0 r}$$

where

$$r = Q - P$$
$$= (1, 2, -3)$$
$$r = \sqrt{1+4+9} = \sqrt{14}$$

$$V = \frac{18 \times 10^{-9}}{4 \times \pi \times \epsilon_0} \times \frac{1}{\sqrt{14}} = 43.23 \text{ volts}$$

c)-



$$V = \int \frac{\rho_L \, dL}{4\pi\epsilon_0 r}$$

$$V = \int \frac{\rho_L}{4\pi\epsilon_0 \times 2} \times (d\rho a_\rho + \rho d\phi a_\phi + dz a_z)$$

$z = 0 \Rightarrow dz = 0$

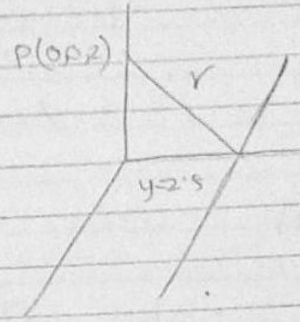
$\phi = \text{constant} \Rightarrow d\phi = 0$

$$V = \int \frac{\rho_L}{4\pi\epsilon_0 \times 2}$$

c)

$$V = \frac{\rho_L}{2\pi\epsilon_0 r}$$

$$r = \sqrt{2^2 + 2.5^2}$$



$$V = \frac{12\times10^{-9}}{2\times\pi\times\epsilon_0\times3.2}$$

$$= 67.4V$$

D: 4.7 :-

each grid = 1mm

a) for 'a'

$$\Delta V = 106 - 104 = 2V$$

$$\Delta L = 2 \text{ grid} = 2mm$$

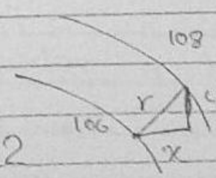
$$\frac{\Delta V}{\Delta L} = 1000 \text{ V/m}$$

but $E = -\frac{\Delta V}{\Delta L} = -1000 \text{ V/m}$

b)

$$\Delta V = 108 - 106 = 2V$$

In figure 4.8: $\Delta x \approx 1.5$ grid $= 1.5mm$



$\Delta y \approx 1.9$ grid $= 1.9mm$

$$r = \sqrt{x^2 + y^2} = \sqrt{1.5^2 + 1.9^2} = 2.42$$

$\Delta V_x \approx 0.9V$; $\Delta V_y \approx 1.3V$

$\frac{\Delta V_x}{\Delta x} = \frac{0.9}{1.5} = 600$; $\frac{\Delta V_y}{\Delta y} = \frac{1.3}{1.9} = 684$

$$E = -\frac{\Delta V_x}{\Delta x}a_x - \frac{\Delta V_y}{\Delta y}a_y = -600a_x - 684a_y$$

c) $\Delta V \approx 0.9$.

$\Delta x \approx 0.7$ grid $= 0.7mm$; $\Delta y \approx 0.75$ grid $= 0.75mm$

$\Delta V_x \approx$

D: 4.8

$$V = \frac{100}{z^2+1} \rho \cos\phi \quad V.$$

$P(\rho = 3, \phi = 60^\circ, z = 2)$.

a) using values of P

$$V = \frac{100}{2^2+1} 3 \times \cos 60 = 30 \text{ volts}.$$

b) $E = -\nabla V$.

as its cylindrical coordinates so.

$$E = -\left[\frac{\partial V}{\partial \rho} a_\rho + \frac{1}{\rho}\frac{\partial V}{\partial \phi} a_\phi + \frac{\partial V}{\partial z} a_z\right].$$

$$= -\left\{\frac{100}{z^2+1}\cos\phi\, a_\rho - \frac{100}{z^2+1}\sin\phi\, a_\phi + \frac{-1}{(z^2+1)^2} \times 2z \times \rho\cos\phi \times 100 a_z\right\}$$

$$= -\left\{10 a_\rho - 17.32 a_\phi - 24 a_z\right\}$$

$$= -10 a_\rho + 17.32 a_\phi + 24 a_z$$

c) $$E = \sqrt{10^2 + 17.32^2 + 24^2}$$

$$= 31.24 \text{ V/m}.$$

d) $$\frac{dV}{dN} = \left.\frac{dV}{dL}\right|_{max}$$

$$\Rightarrow \left|\frac{dV}{dN}\right| = |\nabla V| = 31.2 \text{ V/m},$$

e) ~~a_N~~

$$E = \frac{-dV}{dN} a_N$$

$$\Rightarrow a_N = \frac{-E}{(dV/dN)}$$

$$= 0.32a_\rho - 0.55a_\phi + 0.77a_z.$$

f) $\rho_v = \nabla \cdot D$

$$= \epsilon_0 \nabla \cdot E$$

$$= 224 \text{ pC/m}^3$$

~~where~~

~~$\nabla E = -\cancel{10}$~~

D: 4.9 :-

$$\vec{p} = 3a_x - 2a_y + a_z$$

a) V at P (2,3,4)

$$V = \frac{p \cdot a_r}{4\pi\epsilon_0 r^2}$$ where $r = \sqrt{2^2+3^2+4^2} = \sqrt{29}$.

$$= \frac{3a_x - 2a_y + a_z}{4\pi\epsilon_0 \times 29} \times \frac{2a_x + 3a_y + 4a_z}{\sqrt{29}} \text{ n}$$

$$= \frac{(6-6+4) \times 10^{-9}}{4\pi\epsilon_0 \times 29 \times \sqrt{29}} = 0.23\text{V}.$$

b) converting $\bar{p}$ in spherical coordinates

$$\bar{p} \cdot a_r = 3a_x \cdot a_r - 2a_y \cdot a_r + a_z \cdot a_r$$

$$= 3\sin\theta\cos\phi - 2\sin\theta\sin\phi + \cos\theta$$

$$= 1.372$$

Now using

$$V = \frac{p \cdot a_r}{4\pi\epsilon_0 r^2} = \frac{1.372 \times 10^{-9}}{4 \times \pi \times \epsilon_0 \times 2.5^2}$$

r
$-2r^{-3}$

$= 1.973\ V$

D: 4.10

$\vec{P} = 6a_z\ nC\cdot m$

a) V at $P(r=4,\ \theta=20^\circ,\ \phi=0^\circ)$

as

$$V = \frac{P\cdot a_r}{4\pi\epsilon_0 r^2}$$

$P\cdot a_r = 6a_z\cdot a_r = 6\cos\theta = 5.638\times10^{-9}$

$$V = \frac{5.638\times10^{-9}}{4\pi\epsilon_0\times4^2} = 3.167\ \text{volts}$$

b) -

$E = -\nabla V$

$= \cancel{-\left[\frac{P}{4\pi\epsilon_0}\times -2r^{-1}a_r\right]}$

where

$$V = \frac{6\cos\theta\times10^{-9}}{4\pi\epsilon_0\times r^2}$$

$$\nabla V = \frac{1\times10^{-9}\times6\cos\theta}{4\pi\epsilon_0}(-2r^{-3})a_r + \frac{-6\sin\theta}{4\pi\epsilon_0 r^3}a_\theta$$

$= -1.58\vec{a}_r - 0.288a_\theta$

but

$E = -\nabla V$

$= 1.58\vec{a}_r + 0.288a_r$

Solved by Zaeem. Please report if you find any mistake!

Solved by Zaeem. Please report if you find any mistake!

CHAPTER 7 DRILLS

Solved by Zaeem A. Varaich

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D7.1

(a) $V|_{P(1,2,3)} = \frac{4(2)(3)}{(1)^2+1} = \underline{12\,V}$

As, $\rho_v = -\epsilon\nabla^2 V$, so we first calculate $\nabla^2 V$:

$\nabla V = -8\,\frac{yzx}{(x^2+1)^2} + \frac{4z}{x^2+1} + \frac{4y}{x^2+1}$

$\Rightarrow \nabla^2 V = 32\,\frac{yzx^2}{(x^2+1)^3} - 8\,\frac{yz}{(x^2+1)^2}$

$\Rightarrow \nabla^2 V|_{P(1,2,3)} = 12;$

so, $\rho_v = -\epsilon\nabla^2 V = -\epsilon_o(12) = \underline{-106.25\frac{pC}{m^3}}$

(b) $V|_{P(3,\frac{\pi}{3},2)} = \underline{-22.5\,V}$

As, $\rho_v = -\epsilon\nabla^2 V$, so we first calculate $\nabla^2 V$:

$\nabla^2 V = 20\,\cos(2\,\phi) - 20\,\cos(2\,\phi)$

$\Rightarrow \nabla^2 V|_{P(3,\frac{\pi}{3},2)} = 0;$

so, $\rho_v = -\epsilon\nabla^2 V = -\epsilon_o(0) = \underline{0\,\frac{pC}{m^3}}$

(c) $V|_{P(0.5,45^o,60^o)} = \underline{4\,V}$

As, $\rho_v = -\epsilon\nabla^2 V$, so we first calculate $\nabla^2 V$:

$\nabla^2 V = 4\,\frac{\cos(\phi)}{r^4} - 2\,\frac{\cos(\phi)}{r^4(\sin(\theta))^2}$

$\Rightarrow \nabla^2 V|_{P(0.5,45^o,60^o)} = 0;$

so, $\rho_v = -\epsilon\nabla^2 V = -\epsilon_o(0) = \underline{0\,\frac{pC}{m^3}}$

D7.2

Apply the formulae & concepts to find the answers!

D7.3

(a) The solution to Laplace's equation $\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial}{\partial\rho}\right) = 0$ is:

$V = A\ln\rho + B$

Putting the given values of V & ρ and solving the simultaneous equations, we get:
$A = -73.9$ & $B = 101.28$

so, $V = -73.9\ln\rho + 101.28$

Now, $E = -\nabla V = \frac{1}{\rho}(73.9)\,\mathbf{a}_\rho = \frac{1}{\sqrt{10}}(73.9)\,\mathbf{a}_\rho = 23.36\,\mathbf{a}_\phi$
$(\because \rho = \sqrt{3^2+1^2})$

so, $|E| = \underline{23.36\,\frac{V}{m}}$

(b) The solution to Laplace's equation $\frac{1}{\rho^2}\frac{\partial^2 V}{\partial\phi^2} = 0$ is:

$V = A\phi + B$

Putting the given values of V & ϕ and solving the simultaneous equations, we get:
$A = -85.9$ & $B = 64.9$

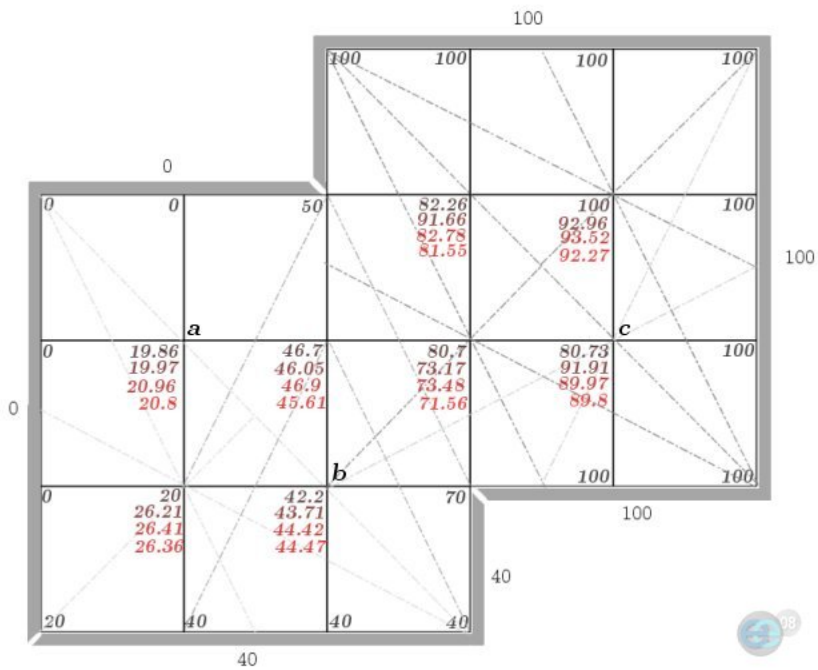
so, $V = -85.9\phi + 64.9$

Now, $E = -\nabla V = \frac{1}{\rho}(85.9)\,\mathbf{a}_\phi = \frac{1}{\sqrt{10}}(85.9)\,\mathbf{a}_\phi = 27.16\,\mathbf{a}_\phi$
$(\because \rho = \sqrt{3^2+1^2})$
so, $|E| = \underline{27.16\,\frac{V}{m}}$

D7.4 & D7.5

Not included in the course!

D7.6



The solution to this problem depends on how you proceed in each iteration and on your initial estimate. *The dotted lines show how the initial estimate was found.*

(a) $\underline{20.8\,V}$

(b) $\underline{44.47\,V}$

(c) $\underline{89.8\,V}$

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CHAPTER 8 DRILLS

(*Upto* **D8.3**)

Solved by Zaeem A. Varaich

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D8.1

(a) Using Δ- form of equation (2), i.e.

$$\Delta H_2 = \frac{I_1 \Delta L_1 \times \mathbf{a}_{\mathrm{R12}}}{4\pi R_{12}^2}$$

Here, $\mathbf{a}_{\mathrm{R12}} = \frac{(4-0)\mathbf{a_x}+(2-0)\mathbf{a_y}+(0-2)\mathbf{a_z}}{\sqrt{4^2+2^2+2^2}} = 0.816\mathbf{a_x} + 0.408\mathbf{a_y} - 0.408\mathbf{a_z}$

$R_{12}^2 = 4^2 + 2^2 + 2^2 = 24$

so, $\Delta H_2 = \frac{I_1 \Delta L_1 \times \mathbf{a}_{\mathrm{R12}}}{4\pi R_{12}^2} = \frac{2\pi \mathbf{a}_z \times (0.816\mathbf{a_x}+0.408\mathbf{a_y}-0.408\mathbf{a_z})}{301.59}\mu = \frac{5.12\mathbf{a_y}-2.56\mathbf{a_x}}{301.59}\mu = \underline{-8.5\mathbf{a_x} + 17.0\mathbf{a_y}\,\frac{nA}{m}}$

(b) As, $\Delta H_2 = \frac{I_1 \Delta L_1 \times \mathbf{a}_{\mathrm{R12}}}{4\pi R_{12}^2}$

Here, $\mathbf{a}_{\mathrm{R12}} = \frac{(4-0)\mathbf{a_x}+(2-2)\mathbf{a_y}+(3-0)\mathbf{a_z}}{\sqrt{4^2+0^2+3^2}} = 0.8\mathbf{a_x} + 0.6\mathbf{a_z}$

$R_{12}^2 = 4^2 + 0^2 + 3^2 = 25$

so, $\Delta H_2 = \frac{I_1 \Delta L_1 \times \mathbf{a}_{\mathrm{R12}}}{4\pi R_{12}^2} = \frac{2\pi \mathbf{a}_z \times (0.8\mathbf{a_x}+0.6\mathbf{a_z})}{100\pi}\mu = \frac{5.02\mathbf{a_y}}{100\pi}\mu = \underline{16\mathbf{a_y}\,\frac{nA}{m}}$

(c) As, $\Delta H_2 = \frac{I_1 \Delta L_1 \times \mathbf{a}_{\mathrm{R12}}}{4\pi R_{12}^2}$

Here, $\mathbf{a}_{\mathrm{R12}} = \frac{(-3-1)\mathbf{a_x}+(-1-2)\mathbf{a_y}+(2-3)\mathbf{a_z}}{\sqrt{4^2+3^2+1^2}} = -0.78\mathbf{a_x} - 0.58\mathbf{a_z} - 0.19\mathbf{a_z}$

$R_{12}^2 = 4^2 + 3^2 + 1^2 = 26$

so, $\Delta H_2 = \frac{I_1 \Delta L_1 \times \mathbf{a}_{\mathrm{R12}}}{4\pi R_{12}^2} = \frac{2\pi(-\mathbf{a_x}+\mathbf{a_y}+\mathbf{2a}_z)\times(-0.78\mathbf{a_x}-0.58\mathbf{a_z}-0.19\mathbf{a_z})}{104\pi}\mu = \frac{(1.96\,\mathbf{a_x}-3.53\,\mathbf{a_y}+2.74\,\mathbf{a_z})\pi}{104\pi}\mu$

$\Rightarrow \Delta H_2 = \underline{18.85\,\mathbf{a_x} - 33.94\,\mathbf{a_y} + 26.40\,\mathbf{a_z}\,\frac{nA}{m}}$

D8.2

Using,

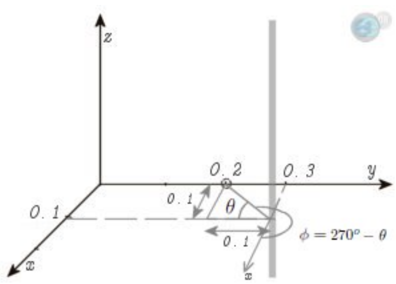
$$\mathbf{H_2} = \frac{I}{2\pi\rho}\mathbf{a}_\phi$$

(a) For $P_A(\sqrt{20},\,0,\,4)$, we have $\rho = \sqrt{20+0} = 4.47$, so:
$\mathbf{H_2} = \frac{15}{2\pi(4.47)}\mathbf{a}_\phi = 0.533\mathbf{a}_\phi = (0.533\times -\sin\phi)\mathbf{a_x} + (0.533\times\cos\phi)\mathbf{a_y}$, where
$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{0}{\sqrt{20}}\right) = 0^o$, so
$\mathbf{H_2} = \underline{0.533\mathbf{a_y}\,\frac{A}{m}}$

(b) For $P_B(2,\,-4,\,4)$, we have $\rho = \sqrt{2^2+4^2} = 4.47$, so:
$\mathbf{H_2} = \frac{15}{2\pi(4.47)}\mathbf{a}_\phi = 0.533\mathbf{a}_\phi = (0.533\times -\sin\phi)\mathbf{a_x} + (0.533\times\cos\phi)\mathbf{a_y}$, where
$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-4}{2}\right) = -63.43^o$, so
$\mathbf{H_2} = (0.533\times 0.89)\mathbf{a_x} + (0.533\times 0.44)\mathbf{a_y} = \underline{0.474\mathbf{a_x} + 0.238\mathbf{a_y}}$

D8.3

(a)



For infinitely long filament;
$\mathbf{H} = \frac{I}{2\pi\rho}\mathbf{a}_\phi$

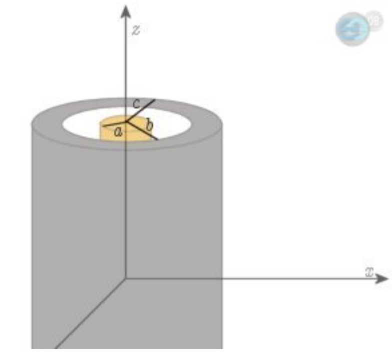
Here,
$\rho = \sqrt{(0.1)^2 + (0.1)^2} = \frac{\sqrt{2}}{10}$

$\mathbf{H} = \frac{I}{2\pi\rho}\mathbf{a}_\phi = \frac{(2.5)}{2\pi(\frac{\sqrt{2}}{10})}\mathbf{a}_\phi = 2.8134\mathbf{a}_\phi = (2.8134\times -\sin\phi)\mathbf{a_x} + (2.8134\times\cos\phi)\mathbf{a_y}$

Now, $\phi = 270^o - \theta = 270^o - \tan^{-1}\left(\frac{0.1}{0.1}\right) = 270^o - 45^o = 225^o$,

so, $\mathbf{H} = (2.8134\times 0.707)\mathbf{a_x} + (2.8134\times -0.707)\mathbf{a_y} = \underline{1.989\mathbf{a_x} - 1.989\mathbf{a_y}\,\frac{A}{m}}$

(b)



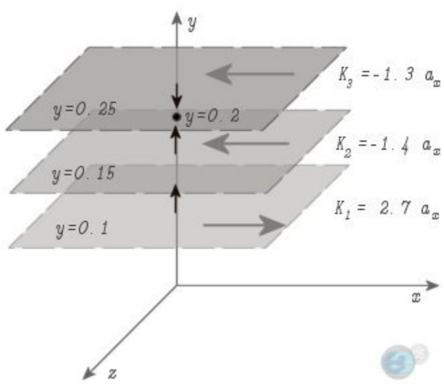
For $\rho < a$,

$\mathbf{H} = \frac{I\rho}{2\pi a^2}\mathbf{a}_\phi = \frac{(2.5)(0.2)}{2\pi(0.3)^2}\mathbf{a}_\phi = 0.884\mathbf{a}_\phi = (0.884 \times -\sin\phi)\mathbf{a_x} + (0.884 \times \cos\phi)\mathbf{a_y}$

Now, $\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{0.2}{0}\right) = 90^o$,

so, $\mathbf{H} = \underline{-0.884\mathbf{a_x}\,\frac{A}{m}}$

(c)



Now,

$\mathbf{H_1} = \frac{1}{2}\mathbf{K_1} \times \mathbf{a_N} = \frac{1}{2}(2.7)\mathbf{a_x} \times \mathbf{a_y} = 1.35\mathbf{a_z}$

$\mathbf{H_2} = \frac{1}{2}\mathbf{K_2} \times \mathbf{a_N} = \frac{1}{2}(-1.4)\mathbf{a_x} \times \mathbf{a_y} = -0.7\mathbf{a_z}$

$\mathbf{H_3} = \frac{1}{2}\mathbf{K_3} \times \mathbf{a_N} = \frac{1}{2}(-1.3)\mathbf{a_x} \times -\mathbf{a_y} = 0.65\mathbf{a_z}$

so, $\mathbf{H} = \mathbf{H_1} + \mathbf{H_2} + \mathbf{H_3} = \underline{1.300\mathbf{a_z}\,\frac{A}{m}}$

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CHAPTER 8 DRILLS

(***D8.4*** *onwards*)

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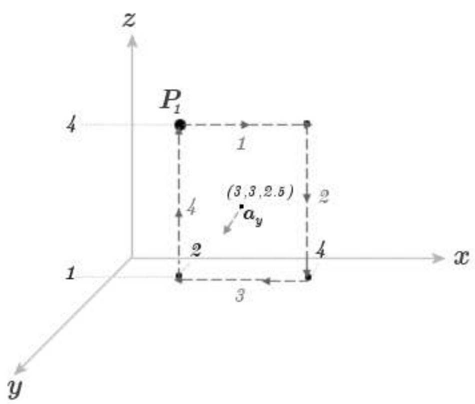
D8.4



(a) For path 1:
$\int(3z\mathbf{a_x} - 2x^3\mathbf{a_z}).(dx\mathbf{a_x} + dy\mathbf{a_y} + dz\mathbf{a_z}) = \int_2^4(3zdx) = 3(4)(4-2) = 24\,A$

For path 2:
$\int(3z\mathbf{a_x} - 2x^3\mathbf{a_z}).(dx\mathbf{a_x} + dy\mathbf{a_y} + dz\mathbf{a_z}) = \int_4^1(-2x^3dz) = -2(4^3)(1-4) = 384\,A$

For path 3:
$\int(3z\mathbf{a_x} - 2x^3\mathbf{a_z}).(dx\mathbf{a_x} + dy\mathbf{a_y} + dz\mathbf{a_z}) = \int_4^2(3zdx) = 3(1)(2-4) = -6\,A$

For path 4:
$\int(3z\mathbf{a_x} - 2x^3\mathbf{a_z}).(dx\mathbf{a_x} + dy\mathbf{a_y} + dz\mathbf{a_z}) = \int_1^4(-2x^3dz) = -2(2^3)(4-1) = -48\,A$

So, $\oint \mathbf{H}.d\mathbf{L} = 24 + 384 - 6 - 48 = \underline{354\,A}$

(b) $\triangle S_N = 3 \times 2 = 6\,m^2$, so

$(\nabla \times \mathbf{H})_y = \frac{\oint \mathbf{H}.d\mathbf{L}}{\triangle S_N} = \frac{354}{6} = \underline{59\frac{A}{m^2}}$

(c) $(\nabla \times \mathbf{H}) = \begin{vmatrix} \mathbf{a_x} & \mathbf{a_y} & \mathbf{a_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = \begin{vmatrix} \mathbf{a_x} & \mathbf{a_y} & \mathbf{a_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3z & 0 & -2x^3 \end{vmatrix} = (0-0)\mathbf{a_x} - (-6x^2-3)\mathbf{a_y} + (0-0)\mathbf{a_z} = (6\text{x}^2+3)\mathbf{a_y}$

At the center, $x = 3,\ z = 2.5$, so

$(\nabla \times \mathbf{H})_y = [6(3)^2 + 3] = \underline{57\frac{A}{m^2}}$

D8.5

(a) $\mathbf{J} = \nabla \times \mathbf{H} = \begin{vmatrix} \mathbf{a_x} & \mathbf{a_y} & \mathbf{a_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = \begin{vmatrix} \mathbf{a_x} & \mathbf{a_y} & \mathbf{a_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & x^2 z & -y^2 x \end{vmatrix} = (-2yx - x^2)\mathbf{a_x} - (-y^2 - 0)\mathbf{a_y} + (2xz - 0)\mathbf{a_z}$

At P: $\mathbf{J} = -16\mathbf{a_x} + 9\mathbf{a_y} + 16\mathbf{a_z}\,\frac{\text{A}}{\text{m}}$

(b) $\mathbf{J} = \nabla \times \mathbf{H} = \left(\frac{1}{\rho}\frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z}\right)\mathbf{a}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho}\right)\mathbf{a}_\phi + \left(\frac{1}{\rho}\frac{\partial(\rho H_\phi)}{\partial \rho} - \frac{1}{\rho}\frac{\partial H_\rho}{\partial \phi}\right)\mathbf{a_z}$

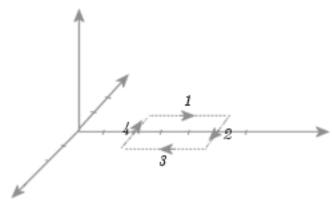
$= \left(\frac{1}{\rho}.0 - 0\right)\mathbf{a}_\rho + (0-0)\,\mathbf{a}_\phi + \left(\frac{1}{\rho}.(2\cos 0.2\phi) - \frac{1}{\rho}\frac{2}{\rho}(-0.2)\sin 0.2\phi\right)\mathbf{a_z}$

At P: $\mathbf{J} = 0.055\mathbf{a_z}\,\frac{\text{A}}{\text{m}^2}$

(c) Using equation (26):
$= \frac{1}{r\sin\theta}(0-0)\,\mathbf{a}_r + \frac{1}{r}(0-0)\,\mathbf{a}_\theta + \frac{1}{r}\left(\frac{\partial}{\partial r}\frac{r}{sin\theta} - 0\right)\mathbf{a}_\phi = \frac{1}{r\sin\theta}\mathbf{a}_\phi$

At P: $\mathbf{J} = \mathbf{a}_\phi\,\frac{\text{A}}{\text{m}^2}$

D8.6



(a) $\oint \mathbf{H}.d\mathbf{L}$:

For path 1:
$\int(6xy\mathbf{a_x} - 3y^2\mathbf{a_y}).(dx\mathbf{a_x} + dy\mathbf{a_y} + dz\mathbf{a_z}) = \int_2^5 (6xydx) = 6(-1)\frac{(5^2-2^2)}{2} = -63\,A$

For path 2:
$\int(6xy\mathbf{a_x} - 3y^2\mathbf{a_y}).(dx\mathbf{a_x} + dy\mathbf{a_y} + dz\mathbf{a_z}) = \int_{-1}^1 (-3y^2dy) = -3\frac{(1^3+1^3)}{3} = -2\,A$

For path 3:
$\int(6xy\mathbf{a_x} - 3y^2\mathbf{a_y}).(dx\mathbf{a_x} + dy\mathbf{a_y} + dz\mathbf{a_z}) = \int_5^2 (6xydx) = 6(1)\frac{(2^2-5^2)}{2} = -63\,A$

For path 4:
$\int(6xy\mathbf{a_x} - 3y^2\mathbf{a_y}).(dx\mathbf{a_x} + dy\mathbf{a_y} + dz\mathbf{a_z}) = \int_1^{-1} (-3y^2dy) = -3\frac{(-1^3-1^3)}{3} = 2\,A$

So, $\oint \mathbf{H}.d\mathbf{L} = -63 - 2 - 63 + 2 = \underline{-126\,A}$

(b) Now, $\int_S (\nabla \times \mathbf{H}).dS$:

$(\nabla \times \mathbf{H}) = \begin{vmatrix} \mathbf{a_x} & \mathbf{a_y} & \mathbf{a_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = \begin{vmatrix} \mathbf{a_x} & \mathbf{a_y} & \mathbf{a_z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy & -3y^2 & 0 \end{vmatrix} = (0-0)\mathbf{a_x} - (0-0)\mathbf{a_y} + (0-6x)\mathbf{a_z} = -6x\mathbf{a_z}$

$(\nabla \times \mathbf{H}).d\mathbf{S} = (-6x\mathbf{a_z})(dydz\mathbf{a_x} + dxdz\mathbf{a_y} + dxdy\mathbf{a_z}) = -6x\,dxdy,$

$\int(\nabla \times \mathbf{H}).dS = -6\int\int x\,dxdy = -6\left(\frac{x^2}{2}\Big|_2^5\right)\left(y|_{-1}^1\right) = -6\frac{(25-4)}{2}(1+1) = \underline{-126\,A}$

D8.7

(a) $H_\phi = \frac{I\rho}{2\pi a^2} = \frac{(20)(0.5m)}{2\pi(1m)^2} = \underline{1592\,\frac{A}{m}}$

(b) $B_\phi = \mu_o H_\phi = 4\pi \times 10^{-7}\frac{(20)(0.8m)}{2\pi(1m)^2} = \underline{3.2\,mT}$

(c) $\phi = \int_S \mathbf{B}.d\mathbf{S} = \int_0^d \int_0^a \frac{I\rho}{2\pi a^2} d\rho dz = \frac{I\mu_O}{2\pi a^2} d \left. \frac{\rho^2}{2} \right|_0^a$, so

$\frac{\phi}{d} = 4\pi \times 10^{-7} \frac{(20)}{4\pi} = \underline{2\,\mu \frac{Wb}{m}}$

(d) $\phi = \int_S \mathbf{B}.d\mathbf{S} = \int_0^d \int_0^{0.5m} \frac{I\rho}{2\pi a^2} d\rho dz = \frac{I\mu_o}{2\pi (1m)^2} d \left. \frac{\rho^2}{2} \right|_0^{0.5m}$, so

$\phi = 4\pi \times 10^{-7} \frac{(20)d(0.5m)^2}{4\pi (1m)^2} = \underline{0.5d\,\mu \frac{Wb}{m}}$

(e) $\phi = \int_S \mathbf{B}.d\mathbf{S} = \int_0^d \int_0^{\infty} \frac{I\rho}{2\pi a^2} d\rho dz = \frac{I\mu_o d}{2\pi (1m)^2} \left. \frac{\rho^2}{2} \right|_0^{\infty}$, so

$\phi = 4\pi \times 10^{-7} \frac{(20)d(\infty)^2}{4\pi (1m)^2} = \underline{\infty \frac{Wb}{m}}$

D8.8

(a) $\mathbf{H} = \frac{I}{2\pi\rho} \mathbf{a}_\phi$, so first we find I:

$I = \int K dN = \int_0^{2\pi} (2.4)(\rho d\phi) = 18.09\, A$

$\mathbf{H} = \frac{18.09}{2\pi\rho} \mathbf{a}_\phi = \underline{\frac{2.88}{\rho} \mathbf{a}_\phi}$

(b) $\mathbf{H} = -\nabla V_m$, so

$\nabla V_m = -\frac{2.88}{(1.5)} \mathbf{a}_\phi = -1.92 \mathbf{a}_\phi$

$\frac{1}{\rho} \frac{\partial V_m}{\partial \phi} = -1.92 \Rightarrow V_m = -2.88 \int_0^{0.6\pi} d\phi = \underline{-5.43V}$

(c) Proceeding similar as above:
$\Rightarrow V_m = -2.88 \int_{2\pi}^{0.6\pi} d\phi = -2.88(0.6 - 2)\pi = \underline{12.66V}$

(d) Similarly,
$\Rightarrow V_m = -2.88 \int_{\pi}^{0.6\pi} d\phi = -2.88(0.6 - 1)\pi = \underline{3.62V}$

(e) Similarly,
$\Rightarrow V_m = \left[-2.88 \int_{-0.6\pi}^{\pi} d\phi \right] + 5 = -14.47 + 5 = \underline{-9.47V}$

D8.9

We know that,

$\mathbf{B} = \mu_o \mathbf{H}$

For solid conductor along z-axis:

$\mathbf{H} = \frac{I\rho}{2\pi a^2} \mathbf{a}_\phi$, so

$\mathbf{B} = \mu_o \frac{\mathrm{I}\rho}{2\pi \mathrm{a}^{\mathbf{2}}} \mathbf{a}_\phi$

Also,

$\mathbf{B} = \nabla \times \mathbf{A}$

Comparing both sides' $\mathbf{a}_\phi$components:

$\mu_o \frac{\mathrm{I}\rho}{2\pi \mathrm{a}^{\mathbf{2}}} = -\frac{\partial \mathrm{A_z}}{\partial \rho} \Rightarrow \mathrm{A_z} = -\int \mu_\mathrm{o} \frac{\mathrm{I}\rho}{2\pi \mathrm{a}^{\mathbf{2}}} \mathrm{d}\rho + \mathcal{C}$

From given conditions, at $\rho = a$ for A_z :

$C = \int \mu_o \frac{\mathrm{I}\rho}{2\pi \mathrm{a}^2} \mathrm{d}\rho + \mu_o \frac{\mathrm{I}\ln \mathbf{5}}{2\pi \mathrm{a}^2} = \mu_o \frac{\mathrm{I}\rho^2}{4\pi \mathrm{a}^2}\Big|_0^a + \mu_o \frac{\mathrm{I}\ln \mathbf{5}}{2\pi \mathrm{a}^2} = 4.218 \times 10^{-7}$

Now,

$A_z = -\mu_o \frac{\mathrm{I}\rho^2}{4\pi \mathrm{a}^2} + (4.39 \times 10^{-7}) = \left[-10^{-7}\frac{\rho^2}{\mathrm{a}^2} + (4.218 \times 10^{-7})\right] I$

(a) At $\rho = 0$

$\mathbf{A} = 0.4218 I \, \mathbf{a_z} \mu \frac{\mathrm{Wb}}{\mathrm{m}}$

(b) At $\rho = 0.25a$

$\mathbf{A} = 0.415 I \, \mathbf{a_z} \mu \frac{\mathrm{Wb}}{\mathrm{m}}$

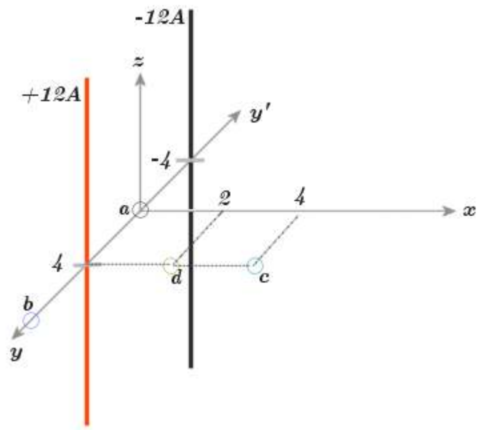
(c) At $\rho = 0.75a$

$\mathbf{A} = 0.365 I \, \mathbf{a_z} \mu \frac{\mathrm{Wb}}{\mathrm{m}}$

(d) At $\rho = a$

$\mathbf{A} = 0.3217 I \, \mathbf{a_z} \mu \frac{\mathrm{Wb}}{\mathrm{m}}$

D8.10



Using Eq. (66) with $b = 4cm$:

$A_z = \frac{\mu_o I}{2\pi} \ln \frac{b}{\rho}$

For red conductor:

$A_{zr} = 2.4 \ln \frac{4cm}{\rho_1} \mu \frac{Wb}{m}$

For black conductor:

$A_{zb} = -2.4 \ln \frac{4cm}{\rho_2} \mu \frac{Wb}{m}$

Also,

$A = A_{zr} + A_{zb}$

(a) $\rho_1 = 4cm = \rho_2$, so $A_{zr} = -A_{zb}$

$A = \underline{0 \frac{Wb}{m}}$

(b) $\rho_1 = 4cm$ and $\rho_2 = 12cm$, so $A_{zb} = 2.63\mu \frac{Wb}{m}$ and $A_{zr} = 0\mu \frac{Wb}{m}$

$A = \underline{2.63\mu\frac{Wb}{m}}$

(c) $\rho_1 = 4cm$ and $\rho_2 = \sqrt{8^2+4^2} = 8.94cm$, so $A_{zr} = 0\mu\frac{Wb}{m}$ and $A_{zb} = 1.93\mu\frac{Wb}{m}$
$A = \underline{1.93\mu\frac{Wb}{m}}$

(d) $\rho_1 = 2cm$ and $\rho_2 = \sqrt{8^2+2^2} = 8.24cm$, so $A_{zr} = 1.66\mu\frac{Wb}{m}$ and $A_{zb} = 1.734\mu\frac{Wb}{m}$
$A = \underline{3.39\mu\frac{Wb}{m}}$

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CHAPTER 9 DRILLS

Solved by Zaeem A. Varaich

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D9.1

(a) $\mathbf{F} = q\mathbf{v} \times \mathbf{B} = \mathrm{qv}\mathbf{a_v} \times \mathbf{B} = 9 \times 10^{-5}(3.3\mathbf{a_x} - 4.5\mathbf{a_y} + 4.65\mathbf{a_z})$

so, $F = \underline{654\mu N}$

(b) $\mathbf{F} = q\mathbf{E} = 18\mathrm{n} \times 10^{3}(-3\mathbf{a_x} + 4\mathbf{a_y} + 6\mathbf{a_z}) = \underline{140.58\mu N}$

(c) $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$

$= (2.97 \times 10^{-4}\mathbf{a_x} - 4.05 \times 10^{-4}\mathbf{a_y} + 4.185 \times 10^{-4}\mathbf{a_z}) + (-5.4 \times 10^{-5}\mathbf{a_x} + 7.2 \times 10^{-5}\mathbf{a_y} + 1.08 \times 10^{-4}\mathbf{a_z})$

$= 2.43 \times 10^{-4}\mathbf{a_x} - 3.3 \times 10^{-4}\mathbf{a_y} + 5.265 \times 10^{-4}\mathbf{a_z}$

so, $F = \underline{664\mu N}$

D9.2

(a) $\mathbf{a_{AB}} = \mathbf{a_x}$, so

$F = -I\oint \mathbf{B} \times d\mathbf{L} = -(12)\oint_1^2(-2\mathbf{a_x} + 3\mathbf{a_y} + 4\mathbf{a_z}) \times \mathbf{a_x}\mathrm{dx}.10^{-3} = -12 \times 10^{-3}\int_1^2 -3\mathbf{a_z} + 4\mathbf{a_y}dx = \underline{-48\mathbf{a_y} + 36\mathbf{a_z}\,mN}$

(b) $\mathbf{a_{AB}} = \frac{2\mathbf{a_x} + 4\mathbf{a_x} + 5\mathbf{a_x}}{3\sqrt{5}}$, so

$\mathbf{F} = -I\oint \mathbf{B} \times d\mathbf{L}$

$= -(12 \times 10^{-3})\left[\int_1^3(-2\mathbf{a_x} + 3\mathbf{a_y} + 4\mathbf{a_z}) \times \mathbf{a_x}\mathrm{dx} + \int_1^5(-2\mathbf{a_x} + 3\mathbf{a_y} + 4\mathbf{a_z}) \times \mathbf{a_y}\mathrm{dy} + \int_1^6(-2\mathbf{a_x} + 3\mathbf{a_y} + 4\mathbf{a_z}) \times \mathbf{a_z}\mathrm{dz}\right]$

$= -(12 \times 10^{-3})\left[\int_1^3(-3\mathbf{a_z} + 4\mathbf{a_y})dx + \int_1^5(-2\mathbf{a_z} - 4\mathbf{a_x})\mathrm{dy} + \int_1^6(2\mathbf{a_y} + 3\mathbf{a_x})\mathrm{dz}\right]$

$= -(12 \times 10^{-3})\left[2(-3\mathbf{a_z} + 4\mathbf{a_y}) + 4(-2\mathbf{a_z} - 4\mathbf{a_x}) + 5(2\mathbf{a_y} + 3\mathbf{a_x})\right]$

$= \underline{12\mathbf{a_x} - 216\mathbf{a_y} + 168\mathbf{a_z}\,mN}$

D9.3

(a) $V = E \times l = 800 \times 1.3c = \underline{10.4\,V}$

(b) $v_d = \mu_e E = 0.13 \times 800 = \underline{104\,\frac{m}{s}}$

(c) $F_t = qvB = (1)(104)(0.07) = \underline{7.28\,\frac{N}{C}}$

(d) $E_t = \frac{F}{q} = \underline{7.28\,\frac{V}{m}}$

(e) $V_H = E_t \times b = 7.28 \times 1.1c = \underline{80.1\,mV}$

D9.4

(a) $d(dF_2) = \mu_o \frac{I_1 I_2}{4\pi R_{12}^2} dL_2 \times (dL_1 \times a_{R12}) = \mu_o \frac{3\times10^{-6}\times3\times10^{-6}}{4\pi(1^2+2^2+2^2)}(-0.5\mathbf{a_x} + 0.4\mathbf{a_y} + 0.3\mathbf{a_z}) \times (\mathbf{a_y} \times \left(\frac{\mathbf{a_x}+2\mathbf{a_y}+2\mathbf{a_z}}{3}\right))$

$= 3.33 \times 10^{-20}(-0.5\mathbf{a_x} + 0.4\mathbf{a_y} + 0.3\mathbf{a_z}) \times (-\mathbf{a_z} + 2\mathbf{a_x}) = 3.33 \times 10^{-20}(-0.4\mathbf{a_x} + 0.1\mathbf{a_y} - 0.8\mathbf{a_z})$

$= \underline{(-1.33\mathbf{a_x} + 0.33\mathbf{a_y} - 2.66\mathbf{a_z}) \times 10^{-20} N}$

(b) $d(dF_1) = \mu_o \frac{I_1 I_2}{4\pi R_{12}^2} dL_1 \times (dL_2 \times a_{R21}) = \mu_o \frac{3\times10^{-6}\times3\times10^{-6}}{4\pi(1^2+2^2+2^2)}(\mathbf{a_y}) \times (-0.5\mathbf{a_x} + 0.4\mathbf{a_y} + 0.3\mathbf{a_z} \times \left(\frac{-\mathbf{a_x}-2\mathbf{a_y}-2\mathbf{a_z}}{3}\right))$

$= 3.33 \times 10^{-20}(\mathbf{a_y}) \times (-0.2\mathbf{a_x} - 1.3\mathbf{a_y} + 1.4\mathbf{a_z}) = 3.33 \times 10^{-20}(0.2\mathbf{a_z} + 1.4\mathbf{a_x})$

$= \underline{(4.67\mathbf{a_x} + 0.66\mathbf{a_z}) \times 10^{-20} N}$

D9.5

(a) $\mathbf{F_A} = I\mathbf{L} \times \mathbf{B} = (0.2)(-4\mathbf{a_x} - 2\mathbf{a_y} + 2\mathbf{a_z}) \times (0.2\mathbf{a_x} - 0.1\mathbf{a_y} + 0.3\mathbf{a_z}) = \underline{-0.08\mathbf{a_x} - 0.32\mathbf{a_y} + 0.16\mathbf{a_z} N}$

(b) $\mathbf{F_A} = -0.08\mathbf{a_x} + 0.32\mathbf{a_y} + 0.16\mathbf{a_z} N$

$\mathbf{F_B} = -0.12\mathbf{a_x} - 0.24\mathbf{a_y} + 0\mathbf{a_z} N$

$\mathbf{F_C} = 0.2\mathbf{a_x} - 0.08\mathbf{a_y} - 0.16\mathbf{a_z} N$

$\mathbf{F} = \mathbf{F_A} + \mathbf{F_B} + \mathbf{F_C} = \underline{0\,N}$

(c) $\mathbf{F_A} = -0.08\mathbf{a_x} + 0.32\mathbf{a_y} + 0.16\mathbf{a_z} N$

$\mathbf{R_A} = \mathbf{a_y} + \mathbf{a_z}$

$\mathbf{T_A} = \mathbf{R_A} \times \mathbf{F_A} = \underline{-0.16\mathbf{a_x} - 0.08\mathbf{a_y} + 0.08\mathbf{a_z}\,N.m}$

(d) It was proved in part (b) that the sum of forces is zero, so:

$\mathbf{T_C} = \mathbf{T_A} = \underline{-0.16\mathbf{a_x} - 0.08\mathbf{a_y} + 0.08\mathbf{a_z}\,N.m}$

D9.6

(a) $M = \frac{B - \mu_o H}{\mu_o} = \frac{\mu H - \mu_o H}{\mu_o} = \underline{1598.87\,A/m}$

(b) $M =$ *(No. of atoms / volume)*$\times$*(Dipole Moment of each atom)*$= 8.3 \times 10^{28} \times 4.5 \times 10^{-27} = \underline{373.5\,A/m}$

(c) $M = \chi_m H = \chi_m \frac{B}{\mu} = 15 \times \frac{300\mu}{(1+\chi_m)\mu_o} = \underline{223.8\,A/m}$

D9.7

(a) $\mathbf{M} = \chi_m \mathbf{H} = \chi_m \frac{\mathbf{B}}{\mu} \Rightarrow \mathbf{B} = \mathbf{M} \times \frac{(1+\chi_m)\mu_o}{\chi_m} = 2.12 \times 10^{-4} z^2 \mathbf{a_x} T$

$\mathbf{J_T} = \nabla \times \frac{\mathbf{B}}{\mu_o} = \nabla \times 168.75 z^2 \mathbf{a_x} = 2 \times 168.75 z \mathbf{a_y}$

so, $J_T|_{z=0.04} = \underline{13.5\, A/m^2}$

(b) $\mathbf{J} = \nabla \times \frac{\mathbf{M}}{\chi_m} = \nabla \times 18.75 z^2 \mathbf{a_x} = 2 \times 18.75 z \mathbf{a_y}$

so, $J\,|_{z=0.04} = \underline{1.5\, A/m^2}$

(c) $\mathbf{J_b} = \nabla \times \mathbf{M} = \nabla \times 150 z^2 \mathbf{a_x} = 2 \times 150 z \mathbf{a_y}$

so, $J_b|_{z=0.04} = \underline{12\, A/m^2}$

D9.8

(a) $|H_{tA}| = |-400\mathbf{a_y} + 500\mathbf{a_z}| = \underline{640.3\, A/m}$

(b) $|H_{NA}| = |300\mathbf{a_x}| = \underline{300\, A/m}$

(c) $\mathbf{H_{t1}} - \mathbf{H_{t2}} = \mathbf{a_{N12}} \times \mathbf{K} \Rightarrow \mathbf{H_{t2}} = \mathbf{H_{t1}} - (\mathbf{a_{N12}} \times \mathbf{K}) = (-400\mathbf{a_y} + 500\mathbf{a_z}) - (200\mathbf{a_y} + 150\mathbf{a_z})$

$\Rightarrow H_{t1} = \underline{694.62\, A/m}$

(d) $H_{N2} = \frac{\mu_1}{\mu_2} H_{N1} = \frac{5}{20} \times 300 = \underline{75\, A/m}$

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CHAPTER 10 DRILLS

Solved by Zaeem A. Varaich

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D10.1

Given:

$\epsilon = 10^{-11}\,\frac{F}{m}$
$\mu = 10^{-5}\,\frac{H}{m}$
$\mathbf{B} = 2 \times 10^{-4} cos 10^5 t\, sin 10^{-3} y\, \mathbf{a_x}\ T$

(a) $\mathbf{B} = \mu \mathbf{H}$, so

$\nabla \times \frac{\mathbf{B}}{\mu} = \epsilon \frac{\partial \mathbf{E}}{\partial t}$ and

$\nabla \times \mathbf{B} = \frac{\partial \mathrm{B_x}}{\partial z} a_y - \frac{\partial \mathrm{B_x}}{\partial y} a_z = 0 - \frac{\partial \mathrm{B_x}}{\partial y} a_z = 2 \times 10^{-7} cos 10^5 t\, cos 10^{-3} y$

$\Rightarrow E = \frac{1}{\mu\epsilon} \int \nabla \times \mathbf{B} dt = \underline{-2 \times 10^4 sin 10^5 t\, cos 10^{-3} y\, a_x\, \frac{V}{m}}$

(b) $\phi = \int_s \mathbf{B}.d\mathbf{S} = \int_\mathbf{s} \mathrm{B_x dydz} = \int_0^{40} \int_0^2 B_x dydz = 2 \times 10^{-7} cos(10^5 \times 10^{-6}) \left(z|_0^2 \right) \frac{\left(-cos(10^{-3}y)|_0^{40}\right)}{10^{-3}} = \underline{0.318\, mWb}$

(Remember to use *radian* mode in calculator while solving this problem)

(c) $\oint \mathbf{E}.d\mathbf{L} = \int_0^2 (-2 \times 10^9 sin 10^5 t\, cos 10^{-3} y) dz - \int_0^2 (-2 \times 10^9 sin 10^5 t\, cos 10^{-3} y) dz$

$= -2 \times 10^4 sin(10^5 10^{-6}) \left(z|_0^2 \right) \left[cos(10^{-3} \times 0) - cos(10^{-3} \times 40) \right] = \underline{-3.19V}$

(Remember to use *radian* mode in calculator while solving this problem)

D10.2

Given:

$d = 7\, cm$
$\mathbf{B} = 0.3\, \mathbf{a_z}\ T$
$\mathbf{v} = 0.1\, \mathbf{a_y} \mathrm{e}^{20\mathrm{y}}\ \frac{m}{s}$
so, $y = \int v dt = \int 0.1\, \mathrm{e}^{20\mathrm{y}} dt = 0.1\, \mathrm{e}^{20\mathrm{y}} t + C$

As, $y = 0$ at $t = 0$
$y = 0.1\, \mathrm{e}^{20\mathrm{y}} t\, \mathbf{a_y} \mathrm{m}$

(a) $v(t = 0) = 0.1\, \mathrm{e}^{20(0)}\ \frac{m}{s} = 0.1\, \frac{m}{s}$

(b) $y(t = 0.1) = 0.1\, \mathrm{e}^{20\mathrm{y}} (0.1)\ \frac{m}{s} \Rightarrow \frac{y}{e^{20y}} = 0.01\, \frac{m}{s}$

(*Solve it using **TABLE** mode in calculator*!)

so, $y(t=0.1) \approx 0.012\,m$

Alternatively, using MAPLE:

$y = -0.05000000000\,LambertW\,(-2.0\,t)$

$y = 0.01295855509\,m = 1.29\,cm$

(c) $v(t=0.1) = 0.1\,e^{20(0.0129)}\ \frac{m}{s} = 0.129\,\frac{m}{s}$

(d) $V_{12} = -Bvd = -0.3 \times 0.129 \times 0.07 = -2.718\,mV$

D10.3

(a) When $J = 0$:

$\nabla \times H = J_d$

$\nabla \times H = -\frac{\partial H_x}{\partial y} = -0.15\cos\left[3.12 \times 3 \times 10^8 t - 3.12y\right] \times -3.12 = 0.468\cos\left[3.12 \times 3 \times 10^8 t - 3.12y\right]$

$|J_d| = |\nabla \times H| = \underline{0.468\,\frac{A}{m^2}}$

(b) When $J = 0$:

$J_d = \nabla \times H = \nabla \times \frac{B}{\mu}$

$\nabla \times B = \frac{\partial H_y}{\partial x} = -0.8\sin\left[1.257 \times 10^{-6} \times 3 \times 10^8 t - 1.257 \times 10^{-6}x\right] \times -1.257 \times 10^{-6}$

$= 1.0056 \times 10^{-6}\sin\left[1.257 \times 10^{-6} \times 3 \times 10^8 t - 1.257 \times 10^{-6}x\right]$

$|J_d| = \left|\nabla \times \frac{B}{\mu}\right| = \underline{0.80023\,\frac{A}{m^2}}$

(c) $\mathbf{D} = \epsilon_r\epsilon_o\mathbf{E}$

$J_d = \frac{\partial \mathbf{D}}{\partial t} = \epsilon_r\epsilon_o\frac{\partial E}{\partial t}$

$= -\epsilon_r\epsilon_o 0.9\sin\left[1.257 \times 10^{-6} \times 3 \times 10^8\mathrm{t} - 1.257 \times 10^{-6}\mathrm{z}\sqrt{5}\right] \times -1.257 \times 10^{-6} \times 3 \times 10^8\mathrm{M} = \underline{0.015025\,\frac{\mathrm{A}}{\mathrm{m}^2}}$

(d) $\mathbf{E} = \frac{\mathbf{J}}{\sigma}$ and $\mathbf{D} = \epsilon_o\mathbf{E}$, so

$|J_d| = \left|\frac{\partial \mathbf{D}}{\partial t}\right| = \epsilon_o\frac{377\mathrm{M}}{\sigma} = \underline{57.55\,\frac{\mathrm{pA}}{\mathrm{m}^2}}$

D10.4

(a) $\nabla.B = 0 \Rightarrow \nabla.\mu H = 0 \Rightarrow \nabla.H = 0$

$\Rightarrow \nabla.(kx\mathbf{a_x} + 10y\mathbf{a_y} - 25z\mathbf{a_z}) = 0$

$\Rightarrow k + 10 - 25 = 0$

$\Rightarrow k = \underline{15A/m^2}$

(b) As $\rho_v = 0$, so $J = \rho_v v = 0$:

$\nabla \times H = \frac{\partial \mathbf{D}}{\partial t} \Rightarrow \mathbf{a_x} = (-4k \times 10^{-9})\mathbf{a_x} \Rightarrow k = \frac{1}{-4 \times 10^{-9}} = \underline{-2.5 \times 10^8 V/(m.s)}$

D10.5

Let $u = 0.64\mathbf{a_x} + 0.6\mathbf{a_y} - 0.48\mathbf{a_z}$

(a) $B_{N1} = B_1.u = 2T$

(b) $B_{t1} = |B_1 \times u| = 3.16T$

(c) $B_{N2} = B_{N1} = 2T$

(d) $B_2 = B_{N2} + B_{t1} = 5.16T$

D10.6

(a) $\rho_s = D_{N1} = |\epsilon_o \epsilon_r \mathbf{E}|_{t=6ns, z=0.3} = 20\epsilon_o\epsilon_r cos(2\times 10^8 t - 2.58z) = 0.806\, nC/m^2$

(b) $\nabla \times E = -\frac{\partial B}{\partial t} = -\frac{\partial \mu H}{\partial t} \Rightarrow H = -\frac{1}{\mu}\int \nabla \times E\, dt$

$= 13.68 \times 10^6 \times \frac{-\cos(2\times 10^8 t - 2.58z)}{2\times 10^8}\mathbf{a_x}\Big|_{t=6ns,\, z=0.3} = -62.3\mathbf{a_x} mA/m$

(c) $H_{t1} = K \times a_N \Rightarrow -62.3\mathbf{a_x} = K \times \mathbf{a_y}$, so $K = -62.3\mathbf{a_z} mA/m$
(Cyclic rule of cross product)

D10.7

(a) $[\rho_{v1}] = 4cos(10^8\pi(t - \frac{R}{v}))$, where $t = 15ns,\ R = 450 - 1.5,\ v = 3\times 10^8 m/s$

$[\rho_{v1}] = 4cos(10^8\pi(t - \frac{R}{v})) = 4\mu$

$[\rho_{v2}] = -4cos(10^8\pi(t - \frac{R}{v}))$, where $t = 15ns,\ R = 450 + 1.5,\ v = 3\times 10^8 m/s$

$[\rho_{v2}] = -4(-1) = 4\mu$

$[\rho_v] = [\rho_{v1}] + [\rho_{v2}] = 8\mu$

Now, considering a unit volume:
$V = \frac{[\rho_v]}{4\pi\epsilon_o R} = 159.77V$

(b) $[\rho_{v1}] = 4cos(10^8\pi(t - \frac{R}{v}))$, where $t = 15ns,\ R = \sqrt{450^2 + 1.5^2} = 450.0025,\ v = 3\times 10^8 m/s$

$[\rho_{v1}] = 4cos(10^8\pi(t - \frac{R}{v})) = -0.01047\mu$

$[\rho_{v2}] = -4cos(10^8\pi(t - \frac{R}{v}))$, where $t = 15ns,\ R = \sqrt{450^2 + 1.5^2} = 450.0025,\ v = 3\times 10^8 m/s$

$[\rho_{v2}] = -4(-1) = 0.01047\mu$

$[\rho_v] = [\rho_{v1}] + [\rho_{v2}] = 0\mu$

Now, considering a unit volume:

$V = \frac{[\rho_v]}{4\pi\epsilon_o R} = 0V$

(c) $[\rho_{v1}] = 4cos(10^8\pi(t - \frac{R}{v}))$, where $t = 15ns,\ R = \sqrt{316.7^2 + 318.2^2} = 449,\ v = 3\times 10^8 m/s$

$[\rho_{v1}] = 4cos(10^8\pi(t - \frac{R}{v})) = 3.46\mu$

$[\rho_{v2}] = -4cos(10^8\pi(t - \frac{R}{v}))$, where $t = 15ns,\ R = \sqrt{319.7^2 + 318.2^2} = 451.06,\ v = 3\times 10^8 m/s$

$[\rho_{v2}] = -4(-1) = 3.58\mu$

$[\rho_v] = [\rho_{v1}] + [\rho_{v2}] = 7.04\mu$

Now, considering a unit volume:
$V = \frac{[\rho_v]}{4\pi\epsilon_o R} = 140.65V$

Please report to the following e-mail, if you find any mistake or typo:

zaeemvaraich@yahoo.com